Sequel Detective

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Faculty Capstone Project

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Problem

- Worksheet-style SQL
- Little investigative context
- Syntax feels like the finish line

Goal

- Case-driven inquiry
- Evidence-backed reasoning
- SQL becomes an investigation tool

The question shifts from

"How do I write a SQL query?"

to

"What does the data prove?"

How The Project Evolved

The core idea stayed the same: students learn SQL by investigating evidence.

1 Analog Classroom Assignment

2 AI-Integrated Capstone Concept

3 Deterministic Web Implementation

This version keeps the investigation, but moves authority into deterministic services.

Learning Experience Is Evidence-Led

Students solve the case by:

- Asking questions
- Testing ideas
- Explaining what the data shows

- **Start with a case question**
- **Explore the database for clues**
- **Use SQL to test each lead**
- **Read results like evidence**
- **Build a conclusion they can defend**

The Student Steps Into The Case

Each screen gives the student a reason to ask the next question.

- Choose the case
- Follow the briefing
- Query for leads
- Build the evidence board
- Test the theory

The App Keeps Evidence Local And Verifiable

The browser guides the investigation, but the backend and database decide what is safe and what is true.

Student Browser

React + Vite + TypeScript
Case screens
Query Lab
Evidence Board

Fastify API

Checks SQL safety
Runs approved queries
Verifies suspects

SQL Server

SequelCityCrimesDB
Case evidence
Verification results

Local Runtime Boundary

Runs on localhost
No direct browser-to-database access
No cloud service required

Students Follow the Evidence

Each query is part of a loop: ask, test, read, revise.

1. Read The Case

Briefing gives the student a reason to investigate.

2. Ask a Data Question

What would prove or weaken this theory?

3. Write a SELECT Query

The student searches for evidence.

4. Read the results

Rows become clues only after interpretation.

5. Revise the theory

The evidence changes what the student believes.

6. Test the Conclusion

The app checks whether the theory matches the case facts.

Blocked queries teach the boundary: explore the evidence, do not alter it.

Demonstration

AI Assisted Engineering Process

Codex implemented scoped work packages;
Gemini and AntiGravity audited against the SSOT.

Codex

- SSOT Creation
- Work Package Creation
- Scoped Implementation
- Mechanical Edits

Gemini / AntiGravity

- Independent audit
- SSOT alignment checks
- Scope and regression review
- Deterministic-boundary validation
- Cross-checking and verification

Human Developer

- Final decisions
- Acceptance review
- Instructional judgment
- Project direction

Successes And Challenges

The project worked best when the learning experience stayed grounded in evidence.

Successes

- Local-first browser runtime
- Deterministic SQL safety boundary
- Playable Case investigation
- Database-backed suspect checks

Challenges

- Turning a static assignment into a guided investigation
- Balancing support with student agency
- Blocking unsafe SQL without blocking exploration
- Hardening local SQL Server setup

Disciplined scope kept the project coherent: deterministic services first, narrative polish second.

Future Plans

Sequel Detective teaches SQL as evidence-based reasoning, not answer hunting.

- Expand the case library
- Persist notebook and case state
- Build Instructor analytics
- Harden the Student Package
- Classroom deployment

Sequel Detective Statistics

Tracked files:	671
Markdown docs:	277
TypeScript files:	103
Documentation lines:	62,153
TypeScript lines:	28,851
SQL script lines:	40,976
Automated test files:	45

Build status: PASSING (web + API)

Test status: PASSING (web 179 + API suite)

All artwork generated by Gemini/Nano Banana

QUESTIONS?

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